KTH • DD2424 • 2026 • GROUP 25

Transfer Learning on the **Oxford-IIIT Pet Dataset**

From Linear Probing to Semi-Supervised Pseudo-Labeling

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37

breeds

01 / THE PROBLEM

Fine-grained classification, limited labels

Can ImageNet features transfer? Can unlabeled data help?

DATASET

Oxford-IIIT Pet

37 breeds (12 cats + 25 dogs)

3,131 training images

549 validation images

3,669 test images

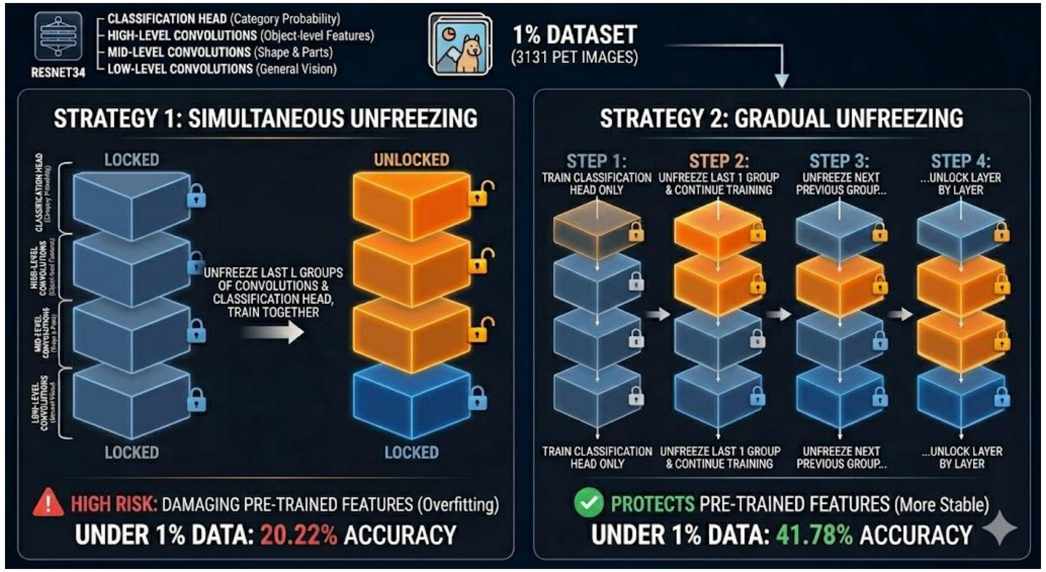
OUR APPROACH

Two investigations

Backbone: ImageNet-pretrained ResNet34

E — **Transfer learning fundamentals**

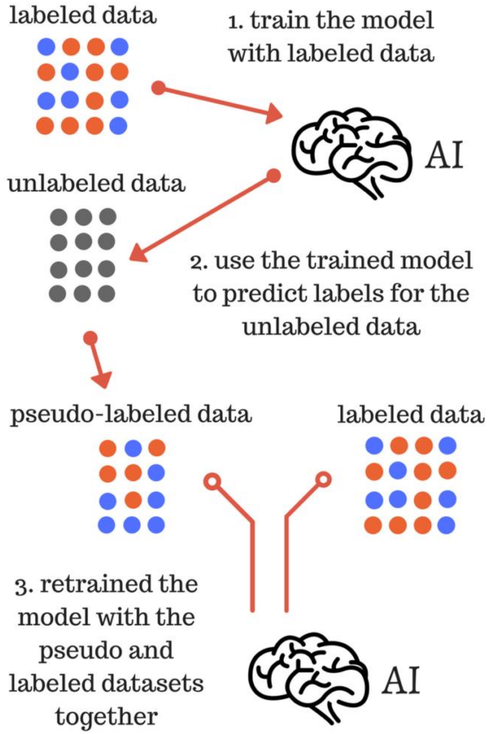
Linear probing • fine-tuning strategies • limited data • class imbalance



This image generated by gemini

B/A — **Semi-supervised extension**

Pseudo-labeling: can unlabeled images compensate for missing labels?



https://ar.inspiredpencil.com/pictures-2023/semi-supervised-learning

Pretrained features go a long way

90.08%

Linear probing

Frozen backbone, only classifier head trained

90.98%

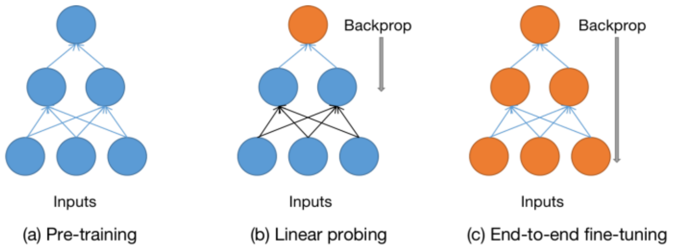
Fine-tuning (Strategy 1, l=3)

Unfreeze last 3 layer groups + head

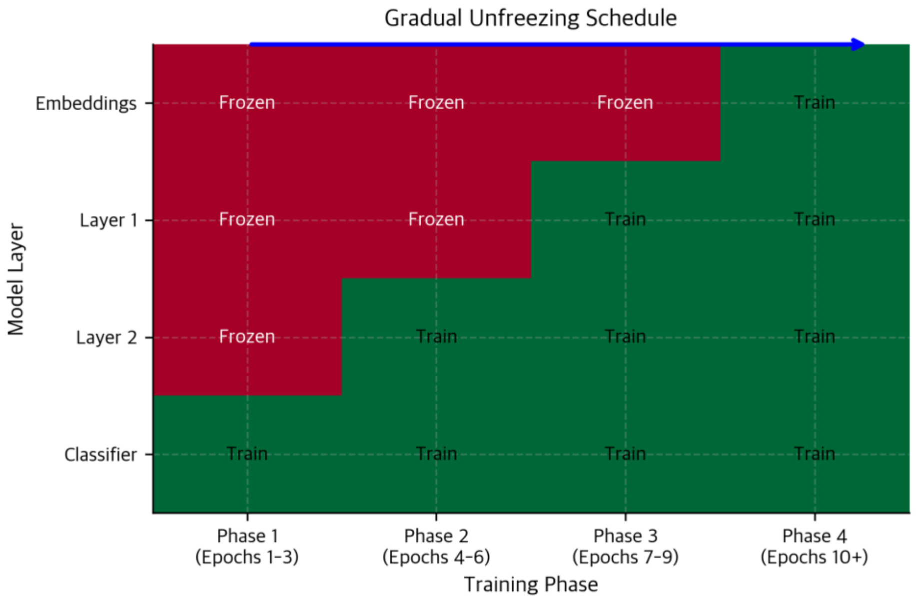
+21.6%

Gradual unfreezing wins at 1% data

41.78% vs 20.22% — preserves pretrained features

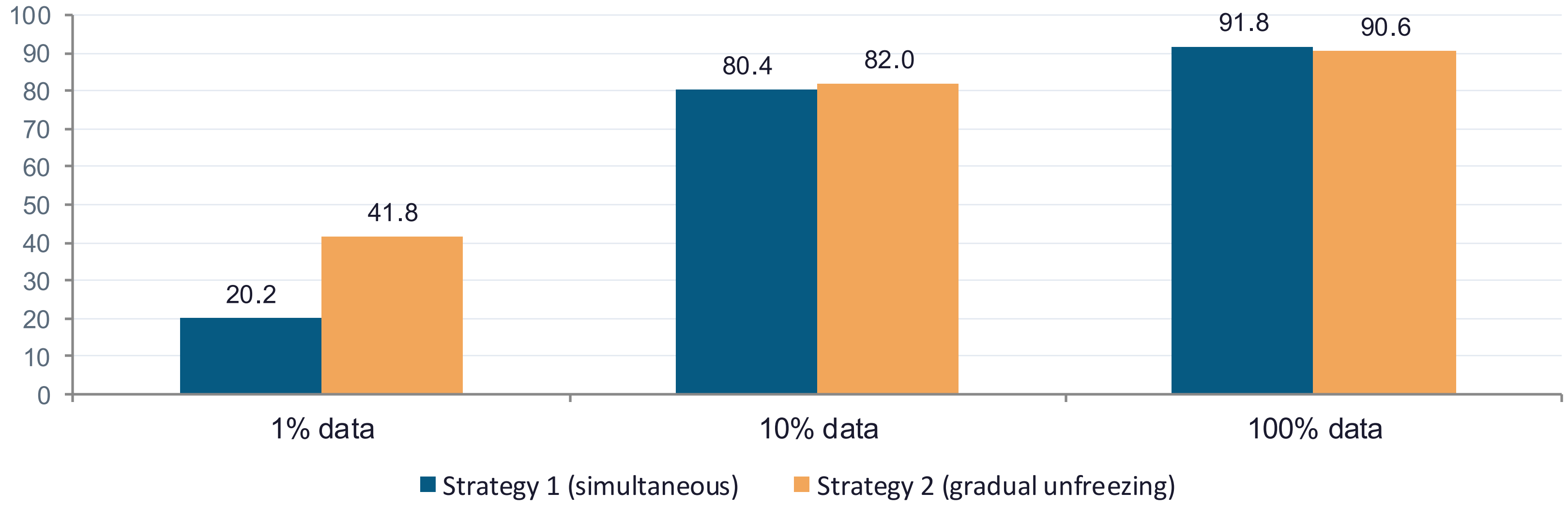


https://www.mdpi.com/remotesensing/remotesensing-14-04824/article_deploy/html/images/remotesensing-14-04824-g004.png



https://cnassets.uk/notebooks/3_catastrophic_forgetting_files/gradual-unfreezing-schedule.png

Strategy comparison across labeled-data fractions

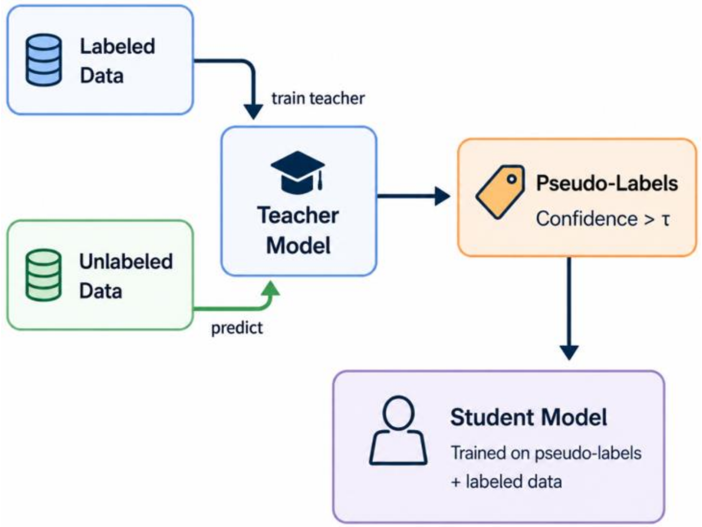


KEY INSIGHT

Gradual unfreezing more than doubles 1% accuracy.

Pretrained representations are preserved instead of destroyed by aggressive updates.

Pseudo-labeling: when does unlabeled data help?



Student vs supervised baseline — at matched label fractions

LABELED	SUPERVISED	+ PSEUDO-LABELING	Δ ACC	PSEUDO-LABELS KEPT
50%	89.07%	**89.59%**	**+0.52**	559 (35.3% accept)
10%	81.14%	81.11%	−0.03	155 (5.5% accept)
1%	40.75%	39.96%	**−0.79**	1 (0.0% accept)

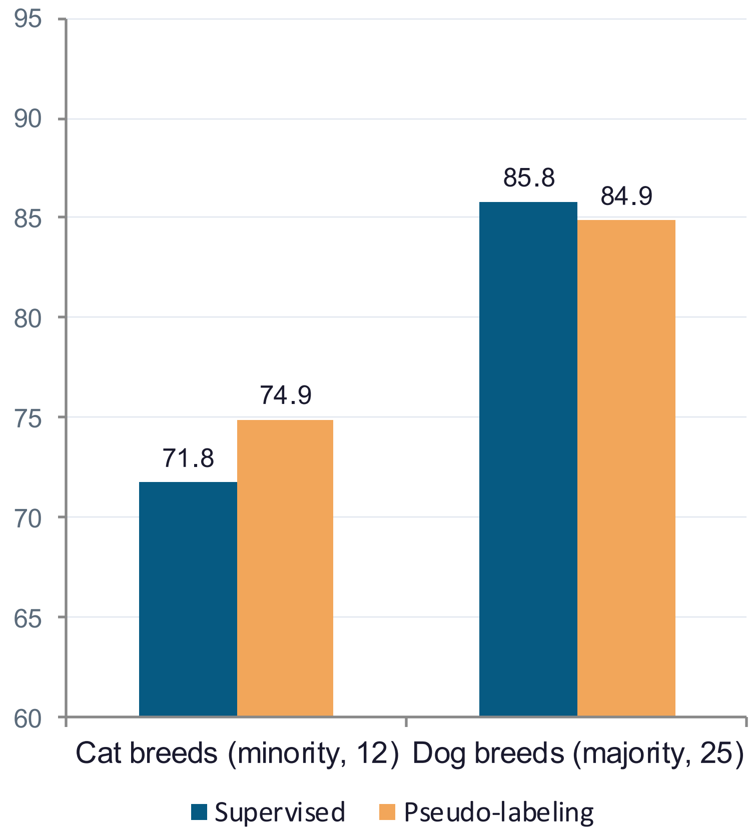
Helpful only when the teacher is reliable AND yields enough high-confidence samples.

At 1% labeled data the teacher accepts a single pseudo-label — too sparse to provide useful supervision.

The student inherits the teacher's blind spots

10% labeled setting · pseudo-label accuracy: 98.71% · but where do the gains go?

Macro F1 by class group



Hard pairs that got worse

Russian Blue → British Shorthair

Supervised errors **22** → With pseudo-labeling **36**

Am. Pit Bull Terrier → Staffordshire Bull Terrier

Supervised errors **23** → With pseudo-labeling **33**

High-confidence ≠ representative.

Accepted pseudo-labels skew toward easy examples — the hard fine-grained boundaries stay unresolved.

https://www.catster.com/wp-content/uploads/2024/06/British-Shorthair-VS-Russian-Blue-Cat-800x571.jpg

https://www.dogster.com/wp-content/uploads/2021/10/American-Staffordshire-Terrier-vs-Pit-Bull-1.png

TAKEAWAYS

What we learned, and what comes next

01

Pretrained features transfer remarkably well

Linear probing alone hits 90% — ImageNet representations capture pet-breed structure.

02

Gradual unfreezing protects what's already learned

Decisive advantage in low-data regimes (41.78% vs 20.22% at 1% labels).

03

Pseudo-labeling is conditional, not universal

Useful at 50% labels; neutral at 10%; harmful at 1%. Depends on teacher quality × accept rate.

04

Easy pseudo-labels can't fix hard fine-grained errors

Next: consistency regularization & class-adaptive thresholds to target the boundaries that matter.

Thank you — questions welcome.